

Hybrid Machine Learning for Uncertainty-Aware Net Load Forecasting in Renewable-Integrated Grids

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Abstract—The integration of renewable energy sources (RES) into power grids has increased variability and uncertainty in net load forecasting. A novel hybrid Bayesian Neural Network-Random Forest (BNN-RF) model is proposed for short-term net load forecasting in high-RES power systems. The model combines BNNs' probabilistic forecasting capabilities with RFs' feature robustness and interpretability. The model achieves a 1.74 % reduction in root mean square error compared to conventional ANN models, providing reliable uncertainty estimates for grid operators managing RES intermittency. The hybrid approach addresses limitations of individual models, highlighting the potential of hybrid machine learning models for improving forecasting accuracy and operational decision-making in renewable-rich power grids.

Index Terms—Net load forecasting, hybrid machine learning, Bayesian neural networks, random forest, renewable energy integration, uncertainty quantification, Artificial Neural Network (ANN), Bayesian Neural Network (BNN), Random Forest (RF)

I. INTRODUCTION

Load forecasting is the action of anticipating the amount of electricity needed at any point in time and how that requirement will impact the overall utility grid. It helps in guaranteeing that there is adequate power production to cater to the needs of the consumer without excessive production of energy, which can contribute to inefficiency and higher cost of running [1]. Another significant consideration in load forecasting is the lead time or forecasting interval, which is the time elapsed between when the forecast is established and when the forecasted load is anticipated to appear. This planning time is necessary for planning and even real-time operational decisions. In day-to-day operation, load forecasting facilitates improved scheduling, load balancing, and resource allocation, which in turn helps in the stability and reliability of the power grid [2]. From a financial perspective, load forecasts improve budget management, lower cost, and facilitate

investment decisions. For generating revenue and operating in the market, load forecasts are of crucial significance in grid structuring and allow for precise forecasting of electrical loads [3], [4]. While load forecasting possesses numerous strengths, it is subject to some limitation. One of the primary challenges is the increasing complexity of contemporary power grids, especially because of the injection of distributed energy resources (DERs) such as solar panels and electric vehicles [5], [6]. These are intermittent and harder to forecast, which makes classic forecasting models more complicated. In addition, weather patterns significantly affect energy demand, and accurate weather forecasting is, therefore, an essential input. While improvements in meteorological technology have enhanced this dimension, there is still much room for further enhancement to extend precision of forecasts [3], [7].

Historical load techniques are long-established tools that have been used to forecast electricity demand in the future. The techniques are mostly based on statistical and time series analysis methods that compare historical load data against each other to detect patterns and trends. Linear regression is among the most popular techniques used, as it develops correlations between electricity load and factors such as ambient temperature, day of the week, season, and time of day. This technique assists utilities in learning how outside variables influence energy use. Moving average methods are also commonly used to level out short-term load data fluctuations, thus identifying potential long-term patterns that could otherwise be lost in daily or weekly changes. A more advanced technique, the Autoregressive Integrated Moving Average (ARIMA) model, incorporates a combination of historical load data and forecast errors in an attempt to increase predictive accuracy. This technique is especially effective for representing non-stationary time series. Exponential smoothing is another well-liked method, which places a higher emphasis on more current observations. This makes the model responsive to changes immediately and captures changing trends in energy consumption. Moreover, end-use models project electricity demand based on known customer behavior and appliance usage habits.

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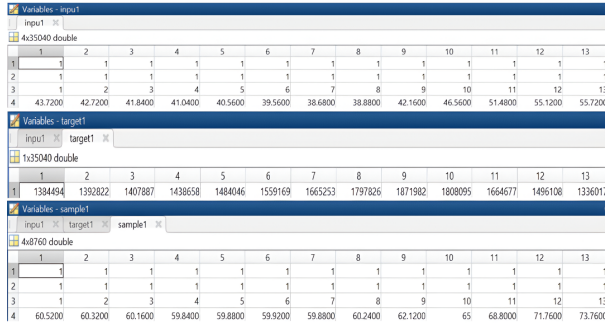


Fig. 1. Input data, Load data and Sample data

Although these classical approaches provide consistent performance in comparatively stable conditions, they are apt to have difficulties adapting to the increasing complexity, variability, and decentralization of contemporary power systems [1], [7], [8].

It is important to create accurate net load forecasting (NLF) models to solve the variability and uncertainty brought about by the growth in the incorporation of renewable energy sources (RES) in power systems [9]. Net load, the difference between overall electricity consumption and RES generation is crucial for the reliability of the grid [1], [10], [11]. Direct NLF, which forecasts net load as a single value, has gained popularity due to its simplicity and available information. The existence of behind-the-meter (BTM) PV systems makes forecasting more complex since their generation is not observable for grid operators. Classical models such as ARIMA tend to be incapable of reflecting nonlinear trends, while machine learning algorithms such as Random Forests (RF) and Bayesian Neural Networks (BNNs) have performed better. On the other hand, machine learning algorithms such as Random Forests (RF) have also exhibited good performance because of their interpretability and robustness [1]. Also, Bayesian Neural Networks (BNNs) provide probabilistic predictions with measurable uncertainty. This is a vital aspect for systems with significant RES penetration [3]. Recent developments suggest that hybrid methods integrating the strengths of a variety of machine learning models can result in enhanced forecasting accuracy. The main contributions of this work are summarized below:

- A novel hybrid BNN-RF model is proposed that combines the uncertainty quantification capabilities of BNNs with the interpretability and robustness of RFs for short-term net load forecasting in power systems with significant RES penetration.
- The forecasting error is reduced by 1.74% in RMSE compared to conventional ANN models, enabling more reliable operational decisions under the intermittency of RES.
- The suggested model interpretability by combining the RF component to identify key weather-driven patterns affecting net load, as well as addressing the black-box nature of traditional neural models.

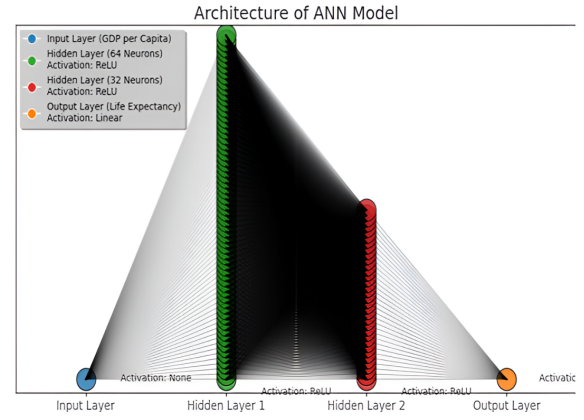


Fig. 2. Architecture of ANN Model

II. PROBLEM DEFINATION

The concise short-term load forecasting (STLF) on an hourly scale plays a crucial role in efficient and cost-effective operation of modern power systems. It allows energy suppliers to make proper generation scheduling decisions, maintain grid stability, smooth export/import of power, and trade energy by minimizing losses and operating costs. This is crucial for a huge power network nation such as India, where electricity demand is highly variable because of geographical climatic variability, socio-economic considerations, and irregular consumption patterns [7]. Fig. 1 presents the input data, load data and sample data. The architecture of ANN model is depicted in Fig. 2.

This work addresses the problem of hourly load forecasting with the use of historical load data. The four years of historical load data consist of 35,064 hourly observations. Every observation contains four main variables like date, hour (whole day), temperature, and load. The target variable is load, while input features are the hour of the day, day of the week, month and temperature—chosen as they are known to impact electricity demand. Generally, Conventional forecasting approaches often fail to effectively model the complex, nonlinear relationships that characterize electricity consumption behaviour, particularly at an hourly resolution. To address this issue, the authors implement and evaluate multiple machine learning techniques known for their flexibility as well as predictive power:

A two-layer feed-forward ANN built using MATLAB's NNtool, trained on transposed column matrices for structured learning. A BNN, also developed in MATLAB, which introduces probabilistic modeling to quantify uncertainty and improve robustness. A RF regression model, implemented using MATLAB's TreeBagger function, leveraging ensemble learning from decision trees to handle multivariate inputs effectively.

In addition, in appreciation of the strength of each model being complementary to the others, a hybrid system is created that weights the BNN and RF models' predictions using a weighted averaging approach [8], [12]. This hybrid approach is intended to leverage the uncertainty-conscious prediction of BNNs and the ensemble robustness of Random Forests to

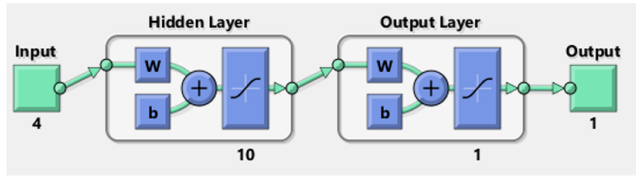


Fig. 3. View of ANN Model in MATLAB

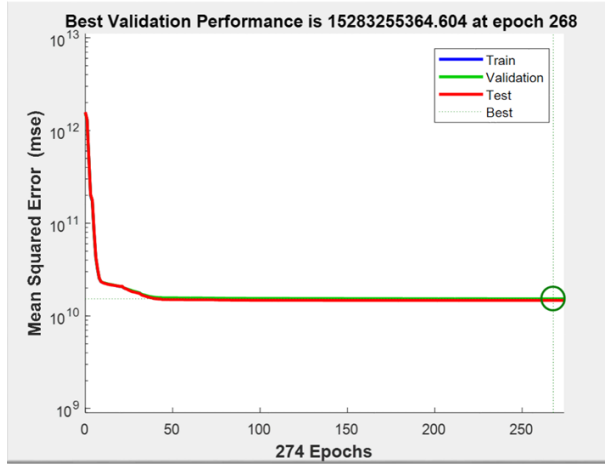


Fig. 4. Performance Plot of ANN Model

produce a more stable and accurate forecasting system. The integration of open-access Indian load data provides real-world relevance and regional context to the research, thereby addressing the lack of open-access high-resolution load data for the Indian case. This also helps increase the reproducibility and availability of the research for potential future studies in energy analytics.

In short, this study will help to advance intelligent energy management systems by testing and combining state-of-the-art machine learning models for one-hour load forecasting. By concentrating on publicly accessible Indian data and taking a comparative as well as hybrid forecasting modeling approach. Many researchers not only pursue enhanced prediction accuracy but also provides a reproducible technique for power system operators as well as researchers in analogous data-scarce regions.

III. ANN BASED APPROACH

The worldwide energy estimate by the International Energy Agency (IEA) of RES capacity will be 460 GW by 2027. Among these, the solar photovoltaic (PV) and wind are major contributors in energy generation across the world [1], [13]. The increasing penetration of RES has emphasised the need for precise NLF to deal with the resulting variability, and uncertainty in power systems [3]. Studies in recent times have proven the efficiency of Artificial Neural Networks (ANNs) to perform short-term NLF (STNLF), especially in micro-neighborhoods with intense RES integration. In comparative studies, ANN models performed better than conventional methods like the ARIMAX model. ANNs can model both linear and nonlinear relations and also resist noisy data [7].

Their architecture normally consists of several hidden neurons and layers as depicted in Fig. 3, and they can be optimized by adjusting hyperparameters like activation functions, learning rates, hidden layer dimensions, and solvers. The mathematical relationship for simple linear regression can be represented as:

$$Y = \beta_0 + \beta_1 * X + \epsilon_0 \quad (1)$$

Where: Y is the dependent variable (or response variable). X is the independent variable (or predictor variable). β_0 is the y-intercept or the constant term. β_1 is the slope coefficient that measures the change in Y for a one-unit change in X. ϵ_0 is the error term or the residuals, which represent the unexplained variation in the dependent variable. The equation for multiple linear regression can be represented as:

$$Y = \beta_0 + \beta_1 * X_1 + \beta_2 * X_2 + \dots + \beta_n * X_n + \epsilon_0 \quad (2)$$

Where: Y is the dependent variable (or response variable). $X_1, X_2, \dots, X_n$ is the independent variable (or predictor variable). β_0 is the y-intercept or the constant term. β_1 are the slope coefficients that measure the change in Y for a one-unit change in $X_1, X_2, \dots, X_n$, respectively. ϵ_0 is the error term or the residuals. The goal of the linear regression model is to estimate the values of the parameters $\beta_0, \beta_1, \beta_2, \dots, \beta_n$ that best fit the data, so that we can make predictions of the dependent variable Y for new values of the independent variables $X_1, X_2, \dots, X_n$. This is typically done by minimizing the sum of squared residuals, which measures the distance between the predicted values and the actual values of the dependent variable.

BNN Based Approach:- Bayesian Neural Networks (BNNs): Extend ANNs by incorporating probabilistic reasoning, offering: uncertainty estimation in forecasts. Robustness to noisy data (common in REST dependent grids). Adaptability to small datasets (useful for microgrid applications). Key Advantages Over ANN that BNN has are as follows: (1) BNN uses prior distributions for weights and updates them via posterior distributions, reducing overfitting and (2) It predicts a distribution of possible net load values (e.g., mean $\pm$ standard deviation) instead of a single point estimate. (3) It handles both model uncertainty and data noise. This approach provides more reliable and interpretable forecasting for STNLF in RES-dominated grids, making it better than the conventional ANN approach for operational decision-making under intermittency [8].

RF Based Approach:- The RF forecasting approach uses several decision trees in combination to make predictions more accurate and robust [14]. The primary advantages of the suggested approach are capturing complex interactions between load, weather, RES generation, and fewer sensitive parameters compared to deep learning models [12], [15], [16]. As a result, the suggested approach offers a robust, interpretable, and scalable alternative for microgrids and distribution networks with high penetration of RES [10], [17]. It ensures effective forecasting and makes it suitable for real-world RES-dominated grid operations.

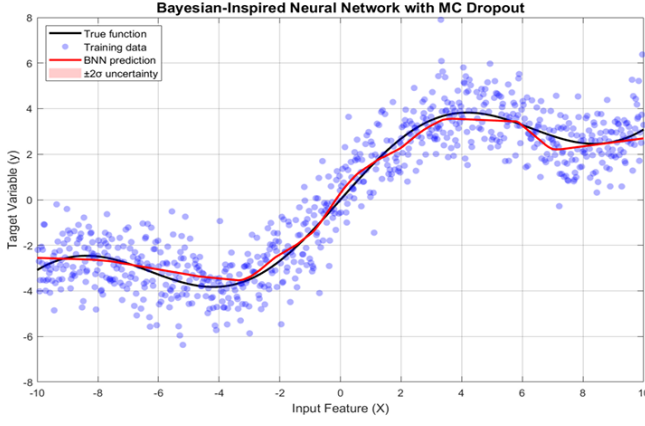


Fig. 5. Regression plot of BNN Model

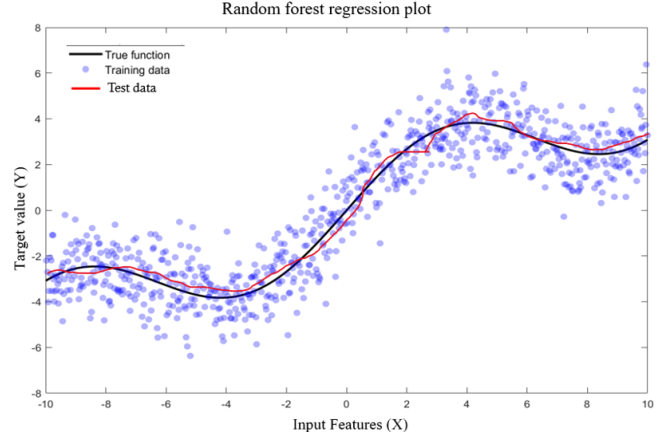


Fig. 6. Regression plot of RF Model

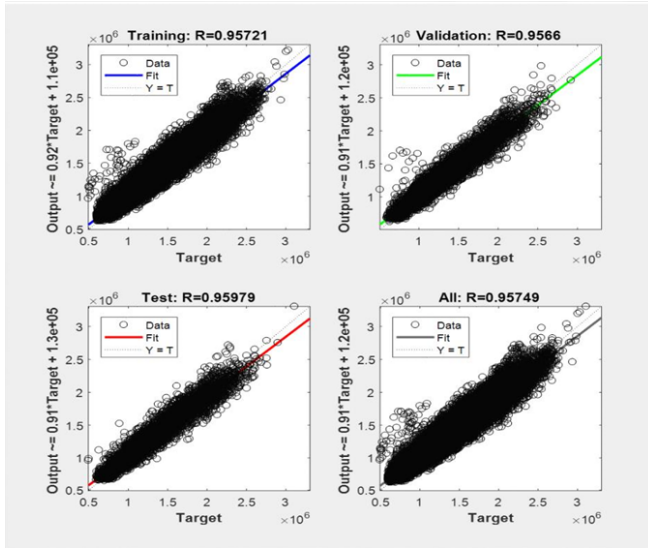


Fig. 7. Regression plot of ANN Model

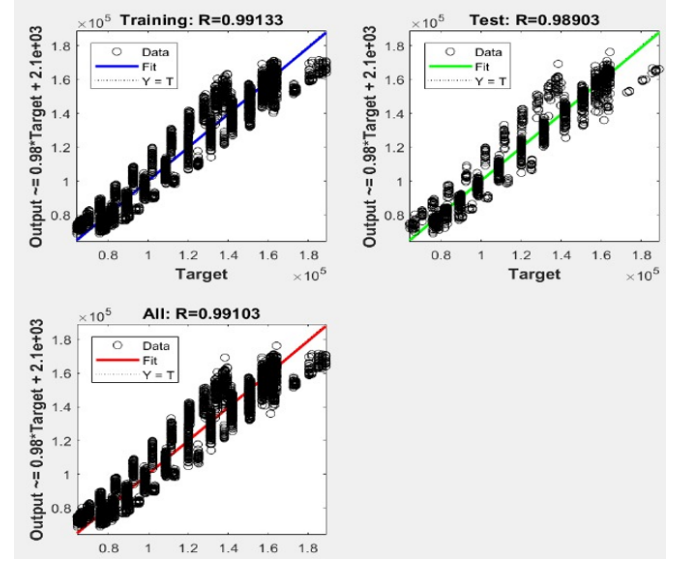


Fig. 8. Regression plot of Hybrid (BNN+RF) Model

IV. PROPOSED HYBRID (BNN+RF) BASED APPROACH

To overcome the above-mentioned issue of STNLF, a hybrid BNN and RF-based forecasting model is used [18], [19]. The BNN provide probabilistic forecasting by directly injecting uncertainty into its predictions. This feature makes it efficient by coping with noisy data and non-linear relationships, which are commonly found in the real world. By using Monte Carlo Dropout both in training and inference steps, the BNN achieves Bayesian inference approximation, which enables it to provide estimates of predictive uncertainty without depending on an advanced sampling approach. This method increases the strength as well as reliability of the model, which is beneficial for applications where knowing the confidence of predictions is as valuable as the predictions themselves. The RF model is strong ensemble learning technique that identifies feature importance as well as models complex nonlinear relationships in data. This design provides it average results across the ensemble. So, it is resistant to overfitting. This mechanism

of averaging facilitates better generalisation, particularly in the context of heterogeneous and high-dimensional datasets like weather variables, load demand, and RES output. Moreover, it also offers a ranking capability of feature importance, which proves useful in knowledge gathering regarding top predictor variables, leading to more interpretable as well as more efficient forecasting models.

Model Architecture and Training: The development of the model started with proper preprocessing of data. Input features are historical load, weather, RES generation and time-based features hour, day and season. All features were normalized with Min-Max scaling in order to keep them compatible for use in neural network models. The data was then divided chronologically into 80% for training and 20% for testing in order to maintain temporal dependencies. The Bayesian Neural Network (BNN) module had a two hidden layer architecture with 64 and 32 neurons each (both employing ReLU activation functions) as depicted in Fig. 3. For adding uncertainty

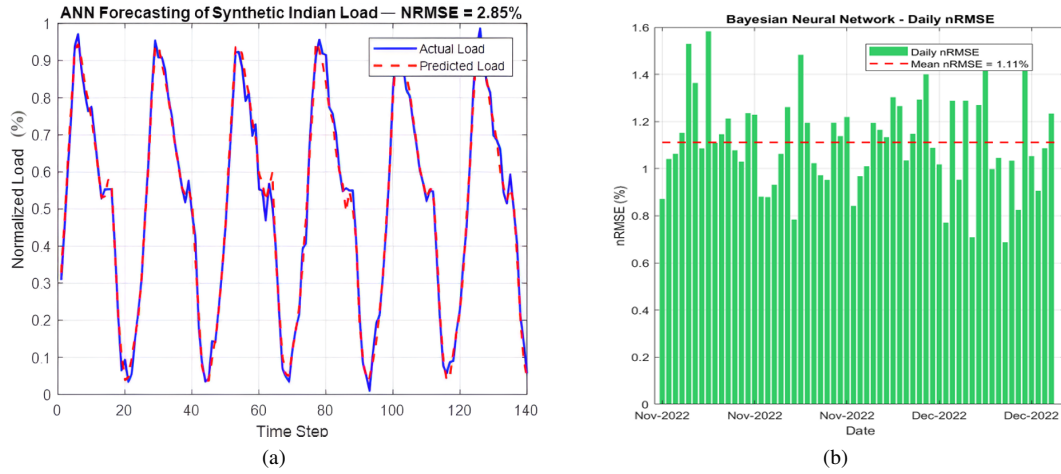


Fig. 9. nRMSE plot for (a) ANN model (b) BNN model

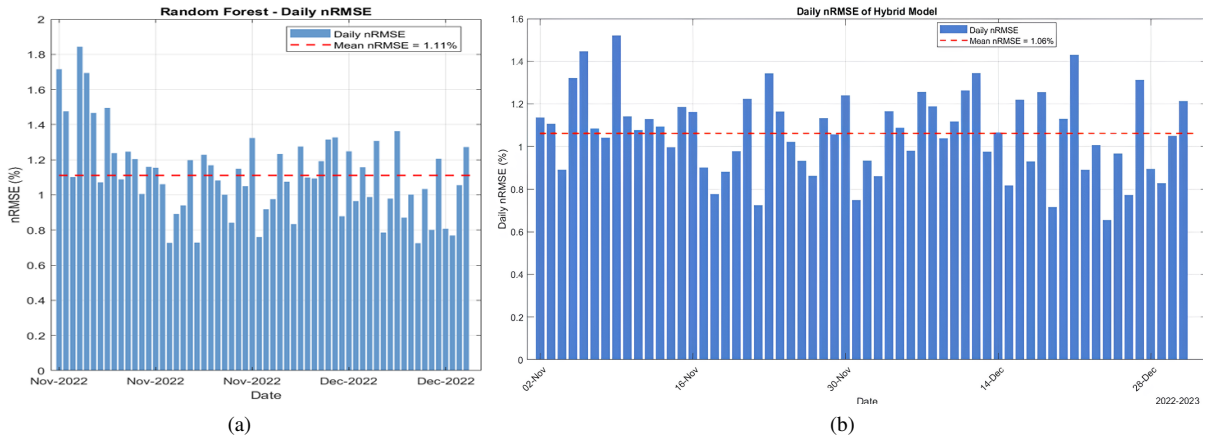


Fig. 10. nRMSE plot for (a) RF model (b) Hybrid (RF+BNN) model

estimation, a dropout layer with a 0.3 rate was used as a Bayesian approximation. The BNN was trained with the Adam optimizer at a learning rate of 0.001 for 100 epochs where early stopping was used to avoid overfitting. Concurrently a RF model was also created. It had 100 decision trees with the maximum number of depths being 15 and the minimum samples split being 5. The feature selection was based on the Gini importance to identify the most important input variables. For prediction, a hybrid integration strategy was adopted with a weighted ensemble of the two models. The ultimate output merged the BNN and RF predictions, giving 60% weightage to BNN and 40% to RF, and producing a prediction enriched with uncertainty bounds. Model performance was assessed with routine regression metrics such as Root Mean Square Error (RMSE), normalized RMSE (nRMSE), and Mean Squared Error (MSE) for accuracy and reliability.

V. RESULTS AND DISCUSSION

This section discusses outcomes acquired via the use of MATLAB/SIMULINK to assess the performance of different machine learning models in comparison to other reported

TABLE I
COMPARATIVE PERFORMANCE ANALYSIS

Training Model	Mean RMSE (in %)	Co-relation value(R)
ANN	2.85	0.957
BNN	1.11	0.969
RF	1.11	0.961
Hybrid(BNN+RF)	1.06	0.991

methodologies. The ANN performance model is predicted in Fig. 4. Furthermore, the regression outcomes of BNN and RF are depicted in Fig. 5 and Fig. 6, respectively. The forecasting is done for 8760 Values on an hourly basis using the ANN, BNN, RF, and Hybrid (BNN+RF) model as depicted in Fig. 7 and Fig. 8, respectively. The regression plot shows the R value for training, validation, test, and all 0.95 for the ANN model in Fig. 9a, around 0.96 for the RF and BNN models in Fig. 9b, and Fig. 10a respectively. The regression plot shows the R value around 0.99 for training, validation, test, and all. This shows that our model is satisfactory and acceptable for the

Hybrid model in Fig. 10(b). The nRMSE for the ANN model is 2.85%, for BNN and RF model it is 1.11%, and for the hybrid it is 1.06% in our case, giving satisfactory and acceptable results, Table I demonstrates comparative performance analysis of ANN, BNN, RF and proposed hybrid (BNN+RF) training models. The mean squared error is improved 100 times in the hybrid model as compared to pure ANN, BNN and RF models. The hybrid model (RF+BNN) outperforms standalone ANN, BNN, and RF models in terms of prediction accuracy by combining learned deep features with robust ensemble decision-making.

VI. CONCLUSION

The proposed hybrid BNN-RF model effectively enhances short-term net load forecasting accuracy in RES-dominated grids. By integrating the probabilistic learning strength of BNNs with the interpretability and robustness of RFs, the suggested approach successfully captures nonlinear dependencies and uncertainty. The 1.75 % reduction in RMSE demonstrates superior performance compared to conventional ANN models. Moreover, the probabilistic forecasts provide valuable uncertainty quantification, which helps grid operators to manage RES intermittency. The developed model thus ensures more reliable operational planning and decision-making. Overall, the BNN-RF hybrid approach represents a promising direction for intelligent forecasting in RES-dominated power grids.

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